Question 1 — Bare-Metal HD44780 LCD Driver

Register configuration:
The LCD control pins are connected to PB0 (RS) and PB1 (EN), so these two pins are configured as outputs using DDRB. The LCD data pins D4–D7 are connected to PD4–PD7, so only the upper four bits of PORTD are used. Bit masking is used so that PD0–PD3 are not changed. The RW pin is connected to ground, so the LCD is used only for writing.

Timing calculations:
Each LCD byte is sent in two 4-bit parts, starting with the upper nibble and then the lower nibble. The EN pin is pulsed each time a nibble is sent. The required LCD waiting times are handled without using delay(): normal commands wait at least 40 µs, while the clear-display command requires at least 1.64 ms. The runtime counter is updated using millis(), which allows the program to continue running without blocking other tasks.

System behavior:
After initialization, the first LCD row displays the name and ID. The second row continuously displays the system runtime in milliseconds. Separate functions are used for sending commands, displaying characters, setting the cursor position, and printing strings.